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QUESTION PAPER

Name of the Examination: Fall 2023-24 Semester – CAT-2

Course Code: CSE1008

Course Title: Theory of computation

Set number: Set-2

Date of Exam: 17/10/2023 (FN)(CI)

Duration: 90 mins

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Obtain the CFG to generate the regular expression $(011 + 1)^* (01)^* (10)$

Q2. Let G be the grammar

$S \rightarrow 0B \mid 1A.$

$A \rightarrow 0 \mid 0S \mid 1AA,$

$B \rightarrow 1 \mid 1S \mid 0BB.$

For the string “00110101”, find (a) the leftmost derivation, (b) the rightmost derivation, and (c) the derivation tree. (10)

Q3. Prove that

a) $S \rightarrow aSbS \mid bSaS \mid \epsilon$ is ambiguous with the string “abab”

b) $S \rightarrow aS \mid aSb \mid x$

$X \rightarrow Xa \mid a$

Where $V = \{S, X\}$ and $T = \{a, b\}$ is ambiguous with the string “abbbb”
(5+5=10)

Q4. Simplify the following grammar by eliminating null production, unit production and useless symbols. (10)

$S \rightarrow a \mid aA \mid B \mid C$

$A \rightarrow aB \mid \epsilon$

$B \rightarrow Aa$

$C \rightarrow cCD$

$D \rightarrow ddd$

Q5. Construct the grammars in Greibach normal form generating the following: (10)

a) $\{wcw^t \mid w \in \{a, b\}^*\}$

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO2	PO1, PO2, PO3	PEO1, PEO4	PSO1, PSO2, PSO3	10
Q2	3	CO2	PO1, PO2, PO3	PEO1, PEO4	PSO1, PSO2, PSO3	10
Q3	3	CO2	PO1, PO2, PO3	PEO1, PEO4	PSO1, PSO2, PSO3	5+5=10
Q4	5	CO3	PO3, PO4, PO5	PEO1, PEO2, PEO3, PEO4	PSO1, PSO2, PSO3	10
Q5	5	CO3	PO3, PO4, PO5	PEO1, PEO2, PEO3, PEO4	PSO1, PSO2, PSO3	10

QUESTION PAPER

Name of the Examination: Fall 2023-24 Semester – CAT-2

Course Code: CSE1008

Course Title: Theory of computation

Set number: Set-4

Date of Exam: 18/10/2023 (An) (O2)

Duration: 90 mins

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Obtain a grammar to generate the following over $\Sigma = \{a, b\}$ (5+5=10)

- a) Set of all strings with exactly one a.
- b) Set of all strings with atleast one a

Q2. For $G = (\{S, A\}, \{a, b\}, S, P)$, where P is

$$S \rightarrow aAS \mid a$$

$$A \rightarrow SbA \mid SS \mid ba$$

Find the leftmost and right most derivation for the string “aabbbaa” and also draw the parse tree. (10)

Q3. Solve the following (5+5=10)

- a) Check whether the given grammar is ambiguous or not by deriving a string

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow id$$

- b) Simplify the grammar by unit and null productions removal

$$S \rightarrow A \mid B$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow abB \mid \epsilon$$

Q4. Transform the CFG into GNF (10)

Given $G = (\{A_1, A_2, A_3\}, \{a, b\}, P, A_1)$ and production P as

$$A_1 \rightarrow A_2 A_3$$

$$A_2 \rightarrow A_3 A_1 \mid b$$

$$A_3 \rightarrow A_1 A_2 \mid a$$

Q5. Show that the language $L = \{a^n \mid n \geq 0\}$ is not a context free. (10)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO2	PO1, PO2, PO3	PEO1, PEO4	PSO1, PSO2, PSO3	10
Q2	3	CO2	PO1, PO2, PO3	PEO1, PEO4	PSO1, PSO2, PSO3	10
Q3	3	CO2	PO1, PO2, PO3	PEO1, PEO4	PSO1, PSO2, PSO3	10
Q4	5	CO3	PO3, PO4, PO5	PEO1, PEO2, PEO3, PEO4	PSO1, PSO2, PSO3	10
Q5	5	CO3	PO3, PO4, PO5	PEO1, PEO2, PEO3, PEO4	PSO1, PSO2, PSO3	10